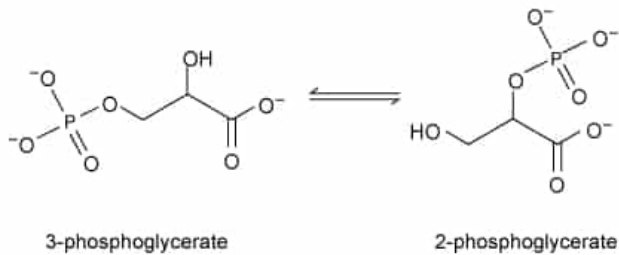


Consider the following reaction:



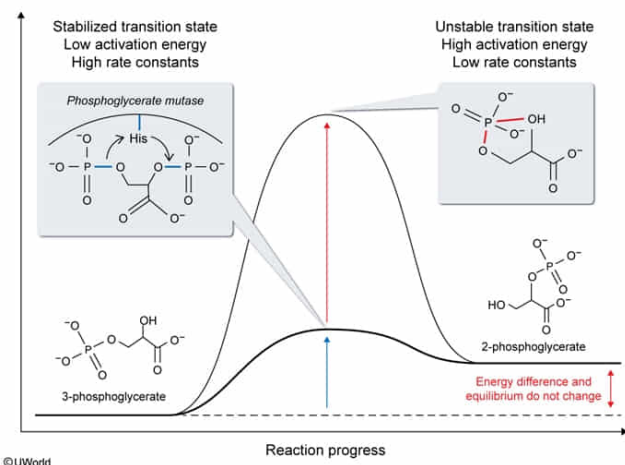
Which parameter would NOT be affected by the addition of phosphoglycerate mutase?

- ☐ A. The stability of the transition state between 3-phosphoglycerate and 2-phosphoglycerate [8%]
- ✓ ☒ B. The equilibrium concentrations of 2-phosphoglycerate and 3-phosphoglycerate [62%]
- ☐ C. The activation energy barrier between 3-phosphoglycerate and 2-phosphoglycerate [8%]
- ☐ D. The rate constant for conversion of 2-phosphoglycerate to 3-phosphoglycerate [20%]

Correct

62% answered correctly

Explanation:



Enzymes are biological **catalysts** that increase the rate of metabolic reactions. Phosphoglycerate mutase is an enzyme that catalyzes the **reversible conversion** of 3-phosphoglycerate (3-PG) to 2-phosphoglycerate (2-PG). The forward reaction (3-PG to 2-PG) occurs in **glycolysis**, and the reverse reaction (2-PG to 3-PG) occurs in **gluconeogenesis**.

Catalysts increase reaction rates by facilitating formation of a **more stable transition state** between reactants and products. This enhanced stability **decreases the activation energy E_a** necessary for the molecules to react, allowing them to reach the energy threshold more often at any given temperature. As a result, the reaction rate increases in both the forward and reverse directions, resulting in an **increased rate constant** as calculated by the [Arrhenius equation](#) (Choices A, C, and D).

Catalysts only stabilize the transition state of a chemical reaction. They do *not* stabilize the reactants or products themselves, and therefore the difference in energy between them (ΔG°) is not affected. The equilibrium constant of a reaction K_{eq} is the ratio of products to reactants at equilibrium, and is related to the standard free energy change by the equation $\Delta G^\circ = -RT \ln(K_{eq})$. Because **ΔG° does not change** in the presence of a catalyst, the equilibrium concentrations of products and reactants also do not change. Therefore, the addition of phosphoglycerate mutase **does not change the equilibrium concentrations** of 2-PG or 3-PG.

Educational objective:

Catalysts, including enzymes, increase reaction rate constants by stabilizing transition states and decreasing activation energy. They do *not* affect the energy states of products or reactants, and therefore they do not affect the equilibrium positions of the reactions they catalyze.

Time spent:0 sec

Subject:
Biochemistry

QID:401759

System:
Enzymes